

Erratum: The Megamaser Cosmology Project. XIII. Combined Hubble constant constraints

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ABSTRACT

In the course of a detailed comparison between the analysis published in our Letter (Pesce et al. 2020a, hereafter P20) and an independent reanalysis carried out by Stiskalek & Desmond (*in prep.*), we have identified several omissions and inaccuracies in P20’s description of the data selection and analysis that we seek to correct with this erratum. The primary correction concerns the use of data quality selection in the maser datasets based on an iterative fit-and-clip procedure used in P20 but not tabulated. We now publish the exact datasets as fit in machine-readable form. We also correct the description of several model parameter prior distributions, and we provide convergence diagnostics for the posterior sampling that were omitted in P20.

1. INPUT DATASETS

Section 2 of P20 stated that its disk fits used the data from the source papers for each of the four galaxies whose distances were re-derived there: Reid et al. (2013) for UGC 3789, Kuo et al. (2013) for NGC 6264, Kuo et al. (2015) for NGC 6323, and Gao et al. (2017) for NGC 5765b¹. However, none of the datasets that P20 fit for the aforementioned four galaxies is identical to the corresponding previously published table.

The dataset differences are primarily the product of data quality selection derived from an iterative fit-and-clip data-vetting process, in which preliminary disk fits were used to identify and remove outlying spots, blinded to knowledge of the distance, at approximately the $\gtrsim 3\sigma$ level in per-coordinate normalized residuals. The physi-

cal rationale for such vetting is that even the “cleanest” megamaser disks can contain emission from non-disk material that a pure disk model should not be forced to accommodate.

The datasets as actually fit by P20 are published with this erratum as Tables 1–4.

1.1. NGC 6264 acceleration uncertainties

The acceleration uncertainties for 33 of the maser spots were inadvertently replaced with values from an internal acceleration analysis, whereby the high-velocity features carry a uniform $0.33 \text{ km s}^{-1} \text{ yr}^{-1}$ uncertainty (the values published in Kuo et al. 2013 range from 0.02 to $0.60 \text{ km s}^{-1} \text{ yr}^{-1}$), and the systemic-feature uncertainties are smaller than the published ones (e.g., $1.07 \pm 0.32 \rightarrow 1.07 \pm 0.11 \text{ km s}^{-1} \text{ yr}^{-1}$). The dataset actually fit in P20 is provided in Table 2; for the sake of reproducibility, the table retains these mis-specified acceleration uncertainties.

Reverting these acceleration uncertainties to their original Kuo et al. (2013) values results in a change in the distance measurement for NGC 6264 from $132.1_{-17.3}^{+21.2} \text{ Mpc}$ to $157.2_{-24.3}^{+31.6}$. However, because this galaxy has such an uncertain distance measurement, its

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¹ We note that the NGC 5765b maser measurements were actually published in Gao et al. (2016), not Gao et al. (2017), and so the citations to Gao et al. (2017) in Section 2.4 of P20 should have pointed to Gao et al. (2016).

impact on the H_0 measurement is modest; reverting the acceleration uncertainties shifts H_0 downward by only $\sim 0.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

1.2. NGC 6323 sky positions and positional uncertainties

In addition to the augmentation of the NGC 6323 dataset described in the caption of Table 3 – i.e., the 68 maser spots in Table 2 of Kuo et al. 2015 plus 19 additional spots from Table 3 of Kuo et al. 2011 – the positions and positional uncertainties in the dataset used in P20 differ from the values published in Kuo et al. (2015). The position differences have median magnitudes in (x, y) of $(2.0, 1.0) \mu\text{as}$ and maximum magnitudes of $(12, 38) \mu\text{as}$, and the positional uncertainties differ by up to $(3.8, 10) \mu\text{as}$. The dataset actually fit in P20 is provided in Table 3

2. MODEL PARAMETER PRIORS

P20 incorrectly described the prior distributions used for some of the error-floor model parameters, and it omitted a description of the prior distributions used for the black hole mass (M) and distance (D) parameters.

2.1. Error-floor parameters

Section 2 of P20 incorrectly stated that all error-floor parameters were assigned broad truncated Gaussian priors, with a mean of $10 \mu\text{as}$ and standard deviation $5 \mu\text{as}$ for the x - and y -position error floors, a mean of 2 km s^{-1} and standard deviation of 1 km s^{-1} for the systemic and high-velocity velocity error floors, and a mean of $0.3 \text{ km s}^{-1} \text{ yr}^{-1}$ and standard deviation of $0.15 \text{ km s}^{-1} \text{ yr}^{-1}$ for the acceleration error floors. However, the priors actually used were galaxy-specific and are listed in Table 5.

2.2. Galaxy distance and black hole mass parameters

P20 did not specify the priors placed on the distance D and black hole mass M in the disk fits, but it referred readers to the model description provided in Pesce et al. (2020b), whose fits applied uniform priors on both parameters. However, the P20 disk fits actually used a joint parameterization – specifically, uniform priors on the pair $\{\log(M/D), \log(M/D^2)\}$ – designed to align the sampled coordinates with the strong degeneracy between M and D and thereby improve sampling efficiency. Because the transformation from $\{\log(M/D), \log(M/D^2)\}$ to $\{\log(D), \log(M)\}$ is linear with unit Jacobian determinant, the sampled prior is equivalent to a prior uniform in $\log(D)$ and $\log(M)$. I.e., $p(D, M) \propto (DM)^{-1}$, restricted to the per-galaxy support given in Table 6.

2.3. Impact

The galaxy-specific prior choices described in this section carry little numerical weight. No distance changes by more than $\sim 1 \text{ Mpc}$ solely on their account, and the net impact on H_0 is at the 0.1σ level.

3. CONVERGENCE DIAGNOSTICS

Each of the four P20 disk fits was run as a single Hamiltonian Monte Carlo chain in PyMC3 (Salvatier et al. 2016), with 10^5 tuning steps followed by 10^5 posterior draws. However, P20 published no convergence diagnostics for these chains.

Table 7 provides within-chain diagnostics computed from the original P20 chains. The tabulated diagnostics are the divergent-transition fraction, the split- $\hat{R}$ of the distance over two and over four chain segments (Gelman & Rubin 1992), and the effective sample sizes (ESSs) of the distance and mass parameters. The summaries published in Table 5 of P20 are the posterior medians and 16th–84th percentile intervals with divergent samples excluded. We note that because the P20 fits were single chains, the statistics in Table 7 are necessarily within-chain measures and cannot by themselves exclude the existence of unexplored, well-separated posterior modes.

4. SUMMARY

The corrections presented here amend the description of the P20 analysis – the provenance and content of its input data tables, its error-floor and distance-mass priors, and its sampling diagnostics – and we provide the material required to reproduce it. None of these corrections alters the analysis that P20 performed or the results it reported.

ACKNOWLEDGMENTS

We are grateful to R. Stiskalek and H. Desmond, whose careful independent reanalysis of the P20 measurements first brought the issues corrected here to our attention.

Table 1. UGC 3789 maser dataset as fit in P20

v	x	σ_x	y	σ_y	a	σ_a	f_a
(km s ⁻¹)	(mas)	(mas)	(mas)	(mas)	(km s ⁻¹ yr ⁻¹)	(km s ⁻¹ yr ⁻¹)	
2472.80	-0.1834	0.0085	-0.1898	0.0085	...	...	0
2477.90	-0.1793	0.0077	-0.1974	0.0211	...	...	0
2562.90	-0.1156	0.0054	-0.1762	0.0279	...	...	0

NOTE—The 153-spot dataset fit in P20. Velocities are in the frame of the source tabulation (Reid et al. 2013); the fitting code added +60 km s⁻¹ to convert to the CMB frame and classified spots with $v < 3000$ km s⁻¹ as blueshifted, $v > 3500$ km s⁻¹ as redshifted, and the remainder as systemic. Three of the 156 maser spots in Table 2 of Reid et al. (2013) were not included in the P20 fit. This table is available in its entirety in machine-readable form.

Table 2. NGC 6264 maser dataset as fit in P20

v	x	σ_x	y	σ_y	a	σ_a	f_a
(km s ⁻¹)	(mas)	(mas)	(mas)	(mas)	(km s ⁻¹ yr ⁻¹)	(km s ⁻¹ yr ⁻¹)	
10918.33	0.3970	0.0056	-0.0330	0.0141	...	...	0
10914.71	0.4030	0.0065	0.0110	0.0154	...	...	0
10911.09	0.4070	0.0048	-0.0050	0.0114	0.11	0.33	1

NOTE—The 61-spot dataset fit in P20. Positions, velocities, and acceleration values are those of Kuo et al. (2013). No velocity-frame conversion was applied, because NGC 6264 lies almost perfectly perpendicular to the CMB dipole direction (with a negligible -0.4 km s⁻¹ projection). Maser spots with $v < 10000$ km s⁻¹ were classified as blueshifted, $v > 10400$ km s⁻¹ as redshifted, and the remainder as systemic. Five of the 66 maser spots in Table 2 of Kuo et al. (2013) were not included in the P20 fit. Furthermore, the acceleration uncertainties of 33 of the 34 retained spots were replaced with values from an internal acceleration analysis (see Section 1.1). This table is available in its entirety in machine-readable form.

Table 3. NGC 6323 maser dataset as fit in P20

v	x	σ_x	y	σ_y	a	σ_a	f_a
(km s ⁻¹)	(mas)	(mas)	(mas)	(mas)	(km s ⁻¹ yr ⁻¹)	(km s ⁻¹ yr ⁻¹)	
8372.94	-0.0180	0.0088	-0.3080	0.0187	...	...	0
8365.82	-0.0200	0.0065	-0.2740	0.0137	...	...	0
8362.25	-0.0380	0.0135	-0.3160	0.0252	...	...	0

NOTE—The 87-spot dataset fit in P20. The fitting code added -33 km s⁻¹ to convert to the CMB frame; spots with $v < 7600$ km s⁻¹ were classified as blueshifted, $v > 8000$ km s⁻¹ as redshifted, and the remainder as systemic. The P20 fit used 87 maser spots: the 68 maser spots of the published fitting set (Table 2 of Kuo et al. 2015) plus 19 additional spots that were members of the published first-epoch catalog (Table 3 of Kuo et al. 2011) but were not retained in the Kuo et al. (2015) catalog. This table is available in its entirety in machine-readable form.

Table 4. NGC 5765b maser dataset as fit in P20

v	x	σ_x	y	σ_y	a	σ_a	f_a
(km s^{-1})	(mas)	(mas)	(mas)	(mas)	($\text{km s}^{-1} \text{ yr}^{-1}$)	($\text{km s}^{-1} \text{ yr}^{-1}$)	
7537.53	-0.3041	0.0141	0.2989	0.0409	...	...	0
7539.30	-0.3399	0.0070	0.3422	0.0202	0.10	0.21	1
7541.08	-0.3319	0.0110	0.3184	0.0319	...	...	0

NOTE—The 169-spot dataset fit in P20. Velocities are in the frame of the source tabulation; the fitting code added $+208 \text{ km s}^{-1}$ to convert to the CMB frame and classified spots with $v < 8000 \text{ km s}^{-1}$ as blueshifted, $v > 8500 \text{ km s}^{-1}$ as redshifted, and the remainder as systemic. The P20 fit used 169 of the 212 maser spots from Table 6 of Gao et al. (2016). 20 systemic spots were excluded for a lack of acceleration measurement, and 23 additional spots were removed through the vetting process. This table is available in its entirety in machine-readable form.

Table 5. Error-floor priors used in the P20 disk fits

Galaxy	σ_x	σ_y	$\sigma_{v,\text{sys}}$	$\sigma_{v,\text{hv}}$	σ_a
	(μas)	(μas)	(km s^{-1})	(km s^{-1})	($\text{km s}^{-1} \text{ yr}^{-1}$)
UGC 3789	$\mathcal{N}(10, 5)$	$\mathcal{N}(10, 5)$	$\mathcal{N}(2, 1)$	$\mathcal{N}(2, 1)$	$\mathcal{N}(0.3, 0.15)$
NGC 6264	$\mathcal{N}(5, 2.5)$	$\mathcal{N}(10, 5)$	$\mathcal{N}(2, 1)$	$\mathcal{N}(2, 1)$	$\mathcal{N}(0.3, 0.15)$
NGC 6323	$\mathcal{N}(10, 5)$	$\mathcal{N}(10, 5)$	$\mathcal{N}(2, 1)$	$\mathcal{N}(2, 1)$	$\mathcal{N}(0.3, 0.15)$
NGC 5765b	$\mathcal{N}(3, 1.5)$	$\mathcal{N}(3, 1.5)$	$\mathcal{N}(1.0, 0.5)$	$\mathcal{N}(1.5, 0.75)$	$\mathcal{N}(0.04, 0.02)$

NOTE— $\mathcal{N}(\mu, \sigma)$ denotes a Gaussian prior with the quoted mean and standard deviation, truncated at zero. Boldface entries indicate galaxy-specific priors that differ from the global priors stated in Section 2 of P20.

Table 6. Distance-mass priors used in the P20 disk fits

Galaxy	D range	M range
	(Mpc)	($10^7 M_\odot$)
UGC 3789	[10, 100]	[0.1, 10]
NGC 6264	[10, 300]	[0.1, 10]
NGC 6323	[0, 1000]	[0, 10]
NGC 5765b	[80, 130]	[2, 20]

NOTE—Within these bounds each fit sampled uniform priors on $\{\log(M/D), \log(M/D^2)\}$, equivalent to $p(D, M) \propto (DM)^{-1}$ (Section 2.2).

Table 7. Within-chain convergence diagnostics for the P20 fits

Galaxy	f_{div} (%)	split- $\hat{R}_2(D)$	split- $\hat{R}_4(D)$	ESS(D)	split- $\hat{R}_2(M)$	split- $\hat{R}_4(M)$	ESS(M)
UGC 3789	4.3	1.0003	1.0034	2346	1.0003	1.0034	2349
NGC 6264	8.1	1.0001	1.0003	8757	1.0002	1.0004	8702
NGC 6323	3.5	1.0030	1.0040	873	1.0030	1.0041	866
NGC 5765b	5.4	1.0007	1.0045	1515	1.0008	1.0044	1462

NOTE—Each P20 fit was computed from a single chain of 10^5 posterior draws (following 10^5 tuning steps). f_{div} is the divergent-transition fraction, split- $\hat{R}_n$ is the Gelman-Rubin statistic computed over n equal chain segments (Gelman & Rubin 1992), and ESS is the effective sample size.

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